

# Cal #36 STANDING Positive-Search — Candidate New SSG Generators v0.1

Lyra (Claude Opus 4.7)

2026-06-04 Thursday ~11:00 EDT

## Cal #36 Positive-Search New SSG Candidates v0.1

### 0. Goal

Per Casey “queue never empties” directive + Cal #36 STANDING RATIFIED Thursday (positive-search shadow of Schur’s lemma): actively hunt for additional substrate Schur generator candidates beyond current 15 cataloged in Lyra SSG Registry. Each candidate identified as Category A (structurally-independent substrate-mechanism FORCING form per Cal #189) or Category B (SSG-7-derived).

Verification gate addressed: Cal #36 STANDING operational across substrate observable landscape.

### 1. Candidate New SSG-16: Substrate Higher Casimir C\_3, C\_4 on D\_IV<sup>5</sup>

Per Lyra Task #189 (completed Sunday morning) + Gate 10 v0.1 substrate-mechanism candidate for T2003:

Candidate substrate source: higher-Casimir operators C\_3 (cubic Casimir) + C\_4 (quartic Casimir) of substrate  $K = SO(5) \times SO(2)$  Lie algebra acting on substrate K-types.

Category: A (structurally-independent — substrate-Lie-algebra higher-Casimir is distinct from quadratic Casimir C\_2 = 6 substrate-spectral sector per SSG-5)

Substrate-mechanism observable candidate: - gen-3 tau mass T2003 =  $g^2 \cdot (2^{C_2} + g)$  substrate-mechanism via higher-Casimir composition (Gate 10 Candidate 1) - Substrate-Pochhammer at higher-Casimir eigenvalues - Substrate-Mehler kernel involving  $\Delta_B^2 + \Delta_B^3$  Bergman Laplacian higher-power compositions

Multi-week explicit content: - C\_3 eigenvalue formula on B\_2 K-types - C\_4 eigenvalue formula on B\_2 K-types - Substrate-Pochhammer factor extension to higher-Casimir Mehler kernel

### 2. Candidate New SSG-17: Substrate Cauchy-Szegő Boundary Kernel

Per Sunday Tier 0 v0.1.6 + Wallach 1976 + FK Ch. XII-XIII Hardy decomposition:

$$\mathcal{H}^2(D_{IV}^5) \cong \mathcal{H}^2(\partial_S D_{IV}^5)$$

Candidate substrate source: Cauchy-Szegő kernel  $S(z, \eta)$  for  $\eta \in \partial_S D_{IV}^5 = (S^4 \times S^1)/Z_2$  — the substrate-Hardy-space boundary reproducing kernel (dual to SSG-7 Bergman kernel bulk reproducing kernel).

Category: A (structurally-independent — substrate-boundary Cauchy-Szegő kernel is distinct from substrate-interior Bergman kernel; substrate-Hardy decomposition is dual K-equivariance class)

Substrate-mechanism observable candidates: - Boundary observables (substrate boundary-projected content) - Casey #12 substrate bulk-boundary projection mechanism explicit derivation - Substrate boundary-charge observables (substrate-Q substrate-charge variants)

Multi-week explicit content: Cauchy-Szegő kernel  $S(z, \eta)$  explicit form per FK Ch. XII-XIII; substrate-mechanism cross-link Gate 2 substrate-vacuum 2-region partition.

### 3. Candidate New SSG-18: Substrate-Coulomb Higher-Multipole Expansion

Per Lyra SSG-10 v0.1 + Hua-coord 3D-projection asymptotic:

Candidate substrate source: substrate-Coulomb beyond  $1/r$  leading asymptotic — higher multipole terms ( $1/r^2$  dipole +  $1/r^3$  quadrupole + ...) substrate-natural decomposition.

Category: B (SSG-7-derived via Hua-coord boundary asymptotic expansion + Bergman kernel matrix elements)

Substrate-mechanism observable candidates: - Atomic spectra fine structure beyond Rydberg (Casey #5 Integer Web instances at hydrogen/atomic scale) - Substrate-Lamb-shift higher-order corrections - Substrate-anomalous magnetic moment  $a_e + a_\mu$  higher-order corrections

Multi-week explicit content: Hua-coord 3D-projection asymptotic expansion in  $1/r^k$  for  $k > 1$ ; substrate-mechanism for atomic multipole moments.

### 4. Candidate New SSG-19: Substrate-Affine $B_2^{(1)}$ Extension

Per Saturday Engine v0.3 +  $B_2$ -affine  $q=2$  Hall algebra K59 RATIFIED:

Candidate substrate source: Affine  $B_2$  ( $B_2^{(1)}$  or  $C_2^{(1)}$ ) substrate extension at substrate-Mersenne SSG-8 + RS coding  $GF(2^g)$  — substrate-affine-Lie-algebra structure.

Category: A (structurally-independent — affine substrate extension is distinct from finite-dimensional  $B_2$  substrate-Lie-algebra; substrate-Affine-Hall-algebra at  $q=2$  substrate primitive)

Substrate-mechanism observable candidates: - Substrate Phase 2 (multi-phase quiver structures) - Substrate macroscopic-substrate-substrate-cognition (Casey #6 Cognition Network) - Substrate  $\Lambda$  exponent  $280 = 2^{\{N_c\}} \cdot n_C \cdot g$  substrate-mechanism via affine-substrate level

Multi-week explicit content:  $B_2^{(1)}$  vs  $C_2^{(1)}$  substrate-Affine PIN per Task #253 + Macdonald 2-parameter ( $q=2, t=\alpha$ ) at affine level.

### 5. Candidate New SSG-20: Substrate Holomorphic Discrete Series K-Type Tower

Per Lyra Wednesday Elie 3742 spinor-tower row + Wallach K-type tower theory + Heckman-Opdam multiplicity:

Candidate substrate source: Full holomorphic discrete series K-type tower on  $D_{IV}^5$  — beyond spinor-tower row  $b/2 = 1/2$  to higher-spin towers ( $b/2 = 1, 3/2, 2, \dots$ ) and beyond half-integer.

Category: B (SSG-7-derived via K-type spectral decomposition; SSG-9 row  $b/2 = 1/2$  is specific tower row, full tower is extension)

Substrate-mechanism observable candidates: - Higher-spin matter K-types (substrate sources for substrate-graviton sector SSG-11  $V_-(2, 0)$ ) - Substrate-Rarita-Schwinger spin-3/2 K-types beyond  $V_-(3/2, 1/2)$  gen-2 muon - Substrate-quark color-triplet K-types via  $SU(3)_C$  extension Cal #194 audit

Multi-week explicit content: full K-type tower enumeration per Heckman-Opdam multiplicity + substrate-mechanism for substrate-massive-spin-2 graviton-sector (SSG-11 v0.9 extension).

### 6. Cal #36 STANDING Positive-Search Summary

5 candidate new SSG generators identified (multi-week investigation): - SSG-16 (Cat A): substrate higher Casimir  $C_3, C_4$  - SSG-17 (Cat A): substrate Cauchy-Szegő boundary kernel - SSG-18 (Cat B): substrate-Coulomb higher-multipole expansion - SSG-19 (Cat A): substrate-Affine  $B_2^{(1)}$  extension - SSG-20 (Cat B): substrate holomorphic discrete series K-type tower

3 NEW Category A candidates added to substrate-mechanism FORCING list (currently 5  $\rightarrow$  8 Category A candidates): - SSG-4 ( $\kappa_{\text{Bergman}}$ ) - SSG-5 (Casimir  $C_2$ ) - SSG-7 (Bergman kernel) - SSG-8 (Mersenne ladder) - SSG-15 (substrate- $\Lambda$ -vacuum) - SSG-16 (higher Casimir) ★ NEW - SSG-17 (Cauchy-Szegő) ★ NEW - SSG-19 ( $B_2^{(1)}$  affine) ★ NEW

## 7. Multi-Week Investigation Path per Candidate

Per Cal #194 WAIT + Cal #189 question-shape + Cal #36 STANDING positive-search:

Each candidate has multi-week explicit derivation path: - SSG-16: explicit higher Casimir eigenvalue formulas + substrate-Mehler kernel extension - SSG-17: Cauchy-Szegő kernel explicit form + substrate-Hardy boundary observable derivation - SSG-18: Hua-coord 3D-projection asymptotic in  $1/r^k$  explicit + atomic-spectra cross-check - SSG-19:  $B_2^{(1)}$  vs  $C_2^{(1)}$  affine PIN + substrate-Affine-Hall-algebra at  $q=2$  explicit - SSG-20: full K-type tower enumeration + Heckman-Opdam multiplicity at general  $(\lambda_1, \lambda_2)$

## 8. Closure v0.1

Cal #36 STANDING positive-search v0.1: 5 candidate new SSG generators identified (3 Category A independent substrate-mechanism FORCING candidates + 2 Category B SSG-7-derived).

Updated SSG candidate landscape: - 15 cataloged (Wednesday end state) - 5 candidate new (Thursday positive-search): SSG-16 + SSG-17 + SSG-18 + SSG-19 + SSG-20 - Total 20 candidate SSGs (15 cataloged + 5 candidate) - 8 Category A independent substrate-mechanism FORCING (current 5 + 3 new candidates)

Per Cal #36 STANDING + Cal #194 WAIT + Cal #189 question-shape: multi-week explicit derivation per candidate gates Category A independence + substrate-mechanism FORCING form determination.

Tier: Cal #36 STANDING positive-search FRAMEWORK OPERATIONAL; 5 candidate new SSGs identified at multi-week investigation level.

— Lyra, Thu 2026-06-04 ~11:00 EDT. Cal #36 positive-search v0.1: 5 candidate new SSGs identified (SSG-16 higher Casimir + SSG-17 Cauchy-Szegő + SSG-18 substrate-Coulomb multipole + SSG-19  $B_2^{(1)}$  affine + SSG-20 K-type tower); 3 Category A added; total 20 candidate SSGs cataloged + candidate; multi-week investigation gates each per Cal #194 WAIT + Cal #189. EOF echo “Cal #36 positive search filed”